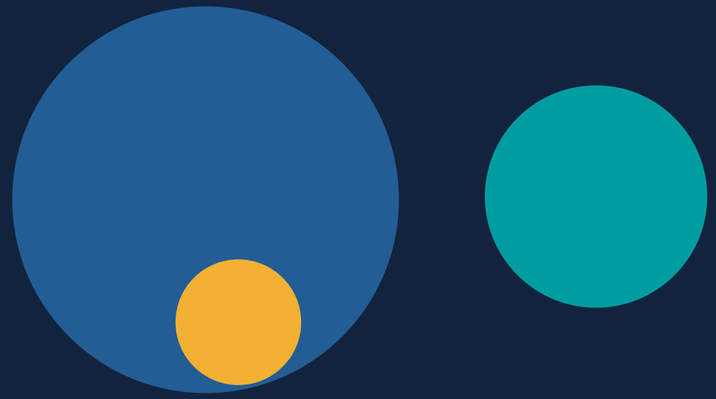
MY TUTOR

An all-in-one learning platform that connects students with personal AI professors for personslised , interactive and effective learning .



TEAM : NEURAL NINJAS
FOUNDER NAME (s) : USMAN RAZA , KHYATI GAUTAM , RAVISH KUMAR PANDEY , MAYANK KUSHWAHA

01 STUDY COMPANION FROM REAL NOTES

Students struggles to access affordable, personalized academic guidance and often rely on multiple platforms to understand concepts , clear doubts , and prepare for exam .

What is the problem ?

Students often struggle to get personalized academic guidance and instant doubt-solving support . they also have to switch multiple platforms for notes , explanations , and exam preparation , making learning time consuming and less effective.

Why does it matters ?

Student lose valuable study time switching between platforms , may spend extra money on tutoring ,and risk falling behind when they cannot get personalized guidance or clear their doubts on time .

Evidence / validation

Students often rely on multiple apps ,You Tube videos ,and private tutors for learning . we can validate this problem through students surveys and feedback on their study challenges and need for personalized AI tutoring.

What happens today?

Students currenty use multiple apps ,You Tube videos , notes , and private tutors to learn and clear doubts ,making studying time-consuming and sometimes expensive .

02 MY TUTOR

What are you building?

My Tutor is an interactive , on-device AI voice tutor and engineering study companion tailored for undergraduate engineering students (B.Tech CSE/IT/ECE/EE/ME).

How does it work?

1. Student uploads notes (PDF, text, or photo read by on-device OCR).
2. Student asks a question.
3. Gemma 4, running locally, answers from the notes and shows the evidence quote; a code check confirms the quote exists in the notes.
4. If the notes don't cover it, the app says so.
5. The app generates practice quizzes and explains wrong answers with hints.

Core features

1.Multi- Language Code Runner
2.Audio Voice Tutoring (Sunao TTS)
3.Multimodel Doubt Input
4.3D Interactive Rubik's Cube Solver

Expected impact

Students get answers they can verify against their own notes, instead of generic AI replies. It works offline, costs nothing per use, and keeps notes on the phone, so it helps students with poor internet or no budget for paid AI tools.

03 Target Users

Primary users

BTech and other college students who revise from their own notes (typed PDFs, text files, or photos of handwritten notes), especially in tier-2 and tier-3 cities.

User needs / pain points

Needs: understand notes quickly, get answers that match their syllabus, and practise with quizzes.
Problems today: generic AI chatbots answer from outside knowledge and can't show what is in the notes; internet is unreliable; paid AI tools are costly; notes are uploaded to cloud servers.

Secondary users / stakeholders

Teachers and mentors who can share notes as practice material, study groups, and colleges that want affordable, privacy-friendly learning tools for students.

Initial audience / reach

Start with BTech students at IIMT University, Meerut, as a pilot group, then expand to other colleges in the region. Add pilot numbers here only when you have real ones (for example, "10 students tested it").

04 The AI Component

Be transparent about the model, why it fits, and how AI is used.

Open-source model / tool

Gemma 4 (E2B / E4B, instruction-tuned) by Google DeepMind, open-weight, Apache 2.0 licence per its Hugging Face page. Run on-device in GGUF format with llama.cpp via the llama_cpp_dart Flutter package.
Link: huggingface.co/google/gemma-4-E2B-it

Why did you choose it?

Small enough to run offline on a student's phone or laptop, so there is no cloud cost and notes stay private. Open weights and a permissive licence fit the open-source goal. Supports 140+ languages, which suits Indian students, and handles long context for notes.

What does the AI do?

Input: the student's notes (PDF, text, or OCR of photos) and a question. Output: an answer built only from the notes, with the supporting quote. If the notes lack the answer, it says "Not found in your notes". It also writes practice MCQs from the notes and explains wrong answers with hints.

Model and implementation

Open-weight model, used as-is (no fine-tuning). Strict prompting so it answers only from the supplied notes, plus selection of the relevant note passages for each question. On-device OCR (Google ML Kit) reads photographed notes. Everything core runs locally.

Validation and safeguards

Code checks that the quoted evidence really appears in the notes, otherwise the answer is replaced by "Not found". Tested on a fixed set of supported and unsupported questions. Notes stay on the device. AI can be wrong, so students are told to verify with textbooks and teachers.

05 Technology Stack

List the technologies you plan to use and what each one does.

Layer	Technology / tool	Purpose / role
Frontend / UI	Flutter (Dart)	Mobile app where students upload notes ,chat with the tutor ,see evidence and take quizzes
Backend / API	fully on-device logic (dart services)	Prompt building,evidence check and quiz generation run inside the app optional internet features
AI / ML	Gemma 4 (E2B/E4B, GGUF)via llama.cpp	Grounded answers,practice questions and feedback generated on -device
Database / RAG	Local files and shared_Preferences	Stores chats, notes and settings on the device .relevant note passages are picked for each question
Hosting / DevOps	GitHub (public repo,open-source licence),Hugging Face (model files)	Version control and the APK release for judges;mode ls downloaded once and then used offline

06 Links & References

Project demo

https://drive.google.com/file/d/18ES5zxqLBIuqwDpW8Q8v28Y6NMJzV9yP/view?usp=drivesdk

Code repository

https://github.com/TricksXTech/My-Tutor

Open-source license : MIT

AI model / tool documentation

Gemma 4 E2B-it model card and licence: huggingface.co/google/gemma-4-E2B-it
Qwen 2.5 1.5B model and licence : https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct-GGUF/resolve/main/qwen2.5-1.5b-instruct-q4_k_m.gguf
Llma 3.2 1B model and licence : https://huggingface.co/bartowski/Llama-3.2-1B-Instruct-GGUF/resolve/main/Llama-3.2-1B-Instruct-Q4_K_M.gguf
Gemma 4 technical report: arxiv.org/abs/2607.02770
llama.cpp (inference engine): github.com/ggml-org/llama.cpp
llama_cpp_dart (Flutter binding): pub.dev/packages/llama_cpp_dart

Research and data sources

Sample notes : e.g. "Semiconductors, own written sample notes (synthetic)"
Google ML Kit text recognition (on-device OCR): pub.dev/packages/google_mlkit_text_recognition
Hacktoberfest 2026 HackDay problem statement PS01 (event booklet)

Contact

Usman Raza, usman.raza.txt@gmail.com, +917817803220
Team : Neural Ninjas, IIMT University, Meerut

THANKYOU